

ECON2050 Tutorial 8

Optimization I

Semester 2 2026

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8 October

Today's Plan

- ▶ Recap: key ideas from Lecture 8
- ▶ Tutorial questions: Q1–Q3

Question 1

Let $g : S \rightarrow \mathbb{R}$ be affine and $h : \mathbb{R} \rightarrow \mathbb{R}$ be convex, where $S \subseteq \mathbb{R}^n$.

(a) Prove that $f = h \circ g$ is convex.

(b) Use this result to show that

$$f_1(x, y) = (y - x)^2 \text{ is convex,}$$

$$f_2(x, y) = -x^2 - 5y^2 - 4xy \text{ is concave.}$$

(c) Find the global minima of f_1 .

(d) Find the global maxima of f_2 .

Recap: Convex and affine functions

Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \rightarrow \mathbb{R}$.

Convex functions

f is **convex** if, for all $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$,

$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

Affine functions

An affine function satisfies

$$g(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) = \lambda g(\mathbf{x}_1) + (1 - \lambda) g(\mathbf{x}_2).$$

Result from last tutorial

$f(x) = x^2$ is strictly convex.

Recap: Critical points and the first-order necessary condition

Critical point

If $f: D \rightarrow \mathbb{R}$ is differentiable, an interior point $\mathbf{x} \in \text{int } D$ is a critical point if

$$\nabla f(\mathbf{x}) = \mathbf{0}.$$

Equivalently,

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = 0, \quad i = 1, \dots, n.$$

First-order necessary condition

If $\mathbf{x}^* \in \text{int } D$ is a local extremum of a differentiable function f , then

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$

Recap: Concavity/convexity gives sufficient conditions

Global extrema from critical points

Let $D \subseteq \mathbb{R}^n$ be convex and let $f: D \rightarrow \mathbb{R}$ be differentiable.

- ▶ If f is concave, every critical point is a global maximum.
- ▶ If f is convex, every critical point is a global minimum.

Thus, for convex or concave objectives, solving

$$\nabla f(\mathbf{x}) = \mathbf{0}$$

can directly identify global extrema.

Knowledge used

- Definition of a convex function:

$$h(\lambda \mathbf{z}_1 + (1 - \lambda) \mathbf{z}_2) \leq \lambda h(\mathbf{z}_1) + (1 - \lambda) h(\mathbf{z}_2).$$

- An affine function satisfies

$$g(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) = \lambda g(\mathbf{x}_1) + (1 - \lambda) g(\mathbf{x}_2).$$

- A critical point of a convex function is a global minimum; a critical point of a concave function is a global maximum.

Tutorial Q1(a): Composition with an affine function

Solution

Take any $\mathbf{x}_1, \mathbf{x}_2 \in S$ and $\lambda \in [0, 1]$. Since g is affine,

$$g(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) = \lambda g(\mathbf{x}_1) + (1 - \lambda) g(\mathbf{x}_2).$$

Therefore,

$$\begin{aligned} f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) &= h(g(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2)) \\ &= h(\lambda g(\mathbf{x}_1) + (1 - \lambda) g(\mathbf{x}_2)). \end{aligned}$$

Because h is convex,

$$h(\lambda g(\mathbf{x}_1) + (1 - \lambda) g(\mathbf{x}_2)) \leq \lambda h(g(\mathbf{x}_1)) + (1 - \lambda) h(g(\mathbf{x}_2)).$$

Hence

$$f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda) f(\mathbf{x}_2).$$

Thus,

$$f = h \circ g \text{ is convex.}$$

The analogous argument gives concavity when h is concave.

Tutorial Q1(b): Convexity and concavity

Solution

For

$$f_1(x, y) = (y - x)^2,$$

write $f_1 = h_1 \circ g_1$, where

$$h_1(z) = z^2 \quad \text{and} \quad g_1(x, y) = x - y.$$

h_1 is convex and g_1 is linear, hence affine. Therefore,

f_1 is convex.

For f_2 ,

$$\begin{aligned} f_2(x, y) &= -x^2 - 5y^2 - 4xy \\ &= -(x + 2y)^2 - y^2. \end{aligned}$$

Both terms are concave, so

f_2 is concave.

Tutorial Q1(c): Global minima of f_1

Solution

The partial derivatives are

$$\frac{\partial f_1}{\partial x} = -2(y - x), \quad \frac{\partial f_1}{\partial y} = 2(y - x).$$

Thus,

$$\nabla f_1(x, y) = \mathbf{0} \iff y = x.$$

So every point on the line $y = x$ is a critical point.

Since f_1 is convex, every critical point is a global minimum. Therefore,

$\{(x, y) \in \mathbb{R}^2 : y = x\}$ is the set of global minima.

Tutorial Q1(d): Global maximum of f_2

Solution

The first-order conditions are

$$\frac{\partial f_2}{\partial x} = -2x - 4y = 0,$$

$$\frac{\partial f_2}{\partial y} = -10y - 4x = 0.$$

Solving gives the unique critical point

$$(x^*, y^*) = (0, 0).$$

Since f_2 is concave, this critical point is a global maximum. Hence

$(0, 0)$ is the unique global maximum of f_2 .

Question 2

Consider

$$f: C \rightarrow \mathbb{R}, \quad f(x, y) = \ln(8y - x^2 + 4x),$$

where C is the domain on which the logarithm is well-defined.

Calculate the critical points of f and determine all local or global extrema.

Recap: Local and global extrema

Global extrema

A point $\mathbf{x}^* \in D$ is a global maximum if

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) \quad \forall \mathbf{x} \in D,$$

and a global minimum if the inequality is reversed.

Local extrema

$\mathbf{x}^*$ is a local maximum if there exists $r > 0$ such that

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) \quad \forall \mathbf{x} \in B_r(\mathbf{x}^*) \cap D.$$

A local minimum is defined analogously.

Every global extremum is a local extremum, but not every local extremum is global.

Recap: Increasing transformations

Preserving extrema

Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and $g : D \rightarrow \mathbb{R}$. If $\mathbf{x}^*$ is a maximum or minimum of g , then it is also a maximum or minimum of $h \circ g$, respectively.

Strictly increasing case

If h is strictly increasing, then the maxima and minima of g and $h \circ g$ coincide.

Recap: Critical points and the first-order necessary condition

Critical point

If $f: D \rightarrow \mathbb{R}$ is differentiable, an interior point $\mathbf{x} \in \text{int } D$ is a critical point if

$$\nabla f(\mathbf{x}) = \mathbf{0}.$$

Equivalently,

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = 0, \quad i = 1, \dots, n.$$

First-order necessary condition

If $\mathbf{x}^* \in \text{int } D$ is a local extremum of a differentiable function f , then

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$

Tutorial Q2: Knowledge used

Knowledge used

- ▶ $\ln z$ is strictly increasing on $z > 0$.
- ▶ A strictly increasing transformation preserves the locations of maxima and minima.
- ▶ First-order necessary condition: an interior local extremum of a differentiable function must be a critical point.
- ▶ Every global extremum is also a local extremum.

Tutorial Q2: Simplify the objective

Solution

Let

$$g(x, y) = 8y - x^2 + 4x, \quad h(z) = \ln z.$$

Then $f = h \circ g$.

Since h is strictly increasing, the local and global maxima/minima of f and g coincide.

The natural domain is

$$C = \{(x, y) \in \mathbb{R}^2 : 8y - x^2 + 4x > 0\}.$$

Equivalently,

$$C = \left\{ (x, y) : y > \frac{(x-2)^2}{8} - \frac{1}{2} \right\},$$

which is an open set.

Tutorial Q2: Critical points and extrema

Solution

For g ,

$$\frac{\partial g}{\partial x} = -2x + 4 = 0 \iff x = 2,$$

while

$$\frac{\partial g}{\partial y} = 8 = 0$$

is impossible.

Therefore, g has no critical points, and hence neither does f .

Because C is open, the first-order necessary condition implies that f has no local extrema. Since every global extremum is local,

f has no local or global maxima or minima.

Question 3

Use the Extreme Value Theorem to show that the following maximization problems have a solution.

(a)

$$\max_{(x,y) \in C} 2xy^2, \quad C = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

(b)

$$\max_{(x,y) \in D} (4x + 7y),$$
$$D = \{(x,y) \in \mathbb{R}^2 : x + 2y \leq 3, x \geq 0, y \geq 0\}.$$

Recap: Extreme Value Theorem

Existence of global extrema

If $D \subseteq \mathbb{R}^n$ is compact and $f: D \rightarrow \mathbb{R}$ is continuous, then f has at least one global maximum and at least one global minimum on D .

What the theorem does and does not say

- ▶ It guarantees existence.
- ▶ It does not identify where the extrema occur.
- ▶ Compactness and continuity are sufficient, not necessary, conditions.

Tutorial Q3: Knowledge used

Knowledge used

- ▶ Extreme Value Theorem: a continuous function on a compact set attains a global maximum and a global minimum.
- ▶ A subset of $\mathbb{R}^n$ is compact if it is closed and bounded.
- ▶ Polynomial and affine functions are continuous.
- ▶ Products, sums, and scalar multiples of continuous functions are continuous.

Tutorial Q3(a): Continuity of the objective

Solution

The objective is

$$f(x, y) = 2xy^2.$$

Using the projection functions $\text{pr}_1(x, y) = x$ and $\text{pr}_2(x, y) = y$,

$$f = 2 (\text{pr}_1)(\text{pr}_2)(\text{pr}_2).$$

The projection functions are continuous, and products and scalar multiples of continuous functions are continuous.

Therefore,

f is continuous on C .

Tutorial Q3(a): Compactness of the constraint set

Solution

The set

$$C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$$

is the closed ball centered at $(0, 0)$ with radius 1.

It is closed because it contains its boundary

$$\{(x, y) : x^2 + y^2 = 1\},$$

and it is bounded because

$$\|(x, y)\| \leq 1 \quad \forall (x, y) \in C.$$

Hence C is compact.

By the Extreme Value Theorem,

$$\max_{(x, y) \in C} 2xy^2 \text{ has a solution.}$$

Tutorial Q3(b): Continuity of the objective

Solution

The objective is

$$g(x, y) = 4x + 7y.$$

Equivalently,

$$g = 4\mathbf{p}r_1 + 7\mathbf{p}r_2.$$

Since the projection functions are continuous, their scalar multiples are continuous, and the sum of continuous functions is continuous.

Thus,

g is continuous on D .

Tutorial Q3(b): Compactness of the constraint set

Solution

The set

$$D = \{(x, y) \in \mathbb{R}^2 : x + 2y \leq 3, x \geq 0, y \geq 0\}$$

is closed because it contains its boundary, and it is bounded; in particular,

$$\|(x, y)\| \leq 3 \quad \forall (x, y) \in D.$$

Therefore, D is compact.

Since g is continuous and D is compact, the Extreme Value Theorem implies

$$\max_{(x,y) \in D} (4x + 7y) \text{ has a solution.}$$

Key takeaways

- ▶ Interior local extrema of differentiable functions must satisfy the first-order conditions.
- ▶ Convexity and concavity turn critical points into global minima or maxima.
- ▶ Strictly increasing transformations preserve the location of extrema.
- ▶ Continuity plus compactness guarantees existence of global extrema.